such a distributed and dynamic environment? That's why we want to keep applications stateless.

Yet, we need to achieve statefulness in the cloud. This section will present the challenges of data services and persistence management in the cloud and describe your options, depending on whether you want to manage a data service yourself or rely on one of the offerings from a cloud provider. Then I'll guide you through setting up a PostgreSQL database instance as a container for your local environment.

5.1.1 Data services in the cloud

Data services are the components of a cloud native architecture designed to be stateful. By designing applications to be stateless, you can limit cloud storage challenges to those few components.

Traditionally, storage was handled by operations engineers and database administrators. But the cloud and DevOps practices enable developers to pick the data service that best fits the application's requirements and deploy it with the same approach used for cloud native applications. Specialists like database administrators are consulted to make the most of the technology that developers have picked, addressing aspects like performance, security, and efficiency. However, the goal is to provide storage and data services on demand, just like you would for cloud native applications, and configure them in a self-service manner.

The difference between application and data services can also be visualized in terms of the three basic building blocks of cloud infrastructure: compute, storage, and network. As shown in figure 5.1, application services use computing and networking resources because they are stateless. On the other hand, data services are stateful and require storage to persist the state.